

Omkar Mahesh

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Education

Vellore Institute of Technology

B.Tech in Computer Science and Engineering (Cloud Computing & Automation)

Sept 2023 – May 2027

CGPA: 8.6/10

Technical Skills

Languages: Python, Go, SQL, JavaScript

Backend & Databases: FastAPI, REST APIs, JWT, SQLAlchemy, Pydantic, PostgreSQL, Redis

Cloud: AWS (EC2, S3, RDS, IAM, VPC, Lambda, DynamoDB, CloudWatch)

DevOps: Linux, Docker, Docker Compose, Nginx, GitHub Actions (CI/CD), Terraform

Tools & Core: Git, GitHub, Postman, Bash, DSA, OOP, DBMS, Operating Systems, Computer Networks

Projects

Meeting Summarizer | *FastAPI, React, PostgreSQL, Redis, Whisper ASR, Docker, Nginx* | [GitHub](#) | [Live](#) **Jul 2026 – Aug 2026**

- Built a full-stack AI meeting assistant with FastAPI and React to automatically transcribe audio and extract actionable summaries, eliminating manual note-taking.
- Engineered an async ASR pipeline with OpenAI and Groq Whisper, delivering sub-2s speech-to-text inference, then structured LLM outputs via Pydantic schemas and JSON mode to extract decisions, action items, and deadlines.
- Architected 6+ REST endpoints with PostgreSQL and Redis caching, securing 100% of data via JWT.
- Containerized and orchestrated 5 services using Docker, Docker Compose, and Nginx reverse proxy, automating CI/CD via GitHub Actions with 18 Pytest suites.

DocuCraft | *FastAPI, React, PostgreSQL, SQLAlchemy, Groq API, Docker, GitHub Actions* | [GitHub](#) | [Live](#) **Jun 2026 – Jul 2026**

- Developed an AI document platform with FastAPI and React for real-time PPT, PDF, and DOCX generation across 5 themes.
- Designed 4+ REST endpoints with FastAPI, SQLAlchemy, and PostgreSQL for auth and per-user document history, securing all routes with JWT authentication, bcrypt hashing, and Pydantic validation.
- Integrated Groq API with structured JSON parsing to generate content across 3 formats and 4–15 slide configs.
- Containerized the backend with Docker and automated GitHub Actions CI/CD, cutting releases to under 3 minutes.

HotReload | *Go, fsnotify, CLI, Process Management* | [GitHub](#) | [Demo](#)

Apr 2026 – May 2026

- Crafted a Go CLI tool automating file-watch, rebuild, and restart cycles, eliminating 100% of manual restarts.
- Configured fsnotify with a 300ms debounce window, reducing redundant build triggers by 80%, across 4 modular components (watcher, builder, runner, engine) with strict separation-of-concerns.
- Hardened process lifecycle with SIGTERM to SIGKILL escalation via Setpgid, guaranteeing zero orphan processes.
- Reduced developer feedback loop by 60% through automated rebuild cycles replacing manual restart workflows.

Certifications

- [AWS Certified Solutions Architect – Associate](#)
- [AWS Certified Cloud Practitioner](#)
- [Microsoft Azure Data Fundamentals \(DP-900\)](#)
- [IBM DevOps Fundamentals](#)
- [Postman API Fundamentals Student Expert](#)

Achievements

- Secured **1st Rank** in the Google Gen AI Exchange Program 2025.
- Contributed to open-source projects during **GSSOC 2024** and **Hacktoberfest**.
- PR & Outreach Lead** at Health O Tech Club, VIT Bhopal, leading public relations and collaboration initiatives.